## SUMMARY

Driving technical innovation in coastal and marine research, with an exposure at a premier marine research center serving the Arabian Gulf for over 40 years, I have established advanced expertise in numerical modeling for environmental assessments and climate-driven ocean dynamics. In my current role as Environmental Modeling Group Lead at KFUPM ARCEMS, I support EIA-related studies for coastal and offshore projects in the Arabian Gulf and liaise closely with client HSE and project teams. I develop and implement environmental and waste management plans, identify key environmental aspects/impacts, and help maintain risk-based registers linked to monitoring and mitigation measures. I plan and execute environmental monitoring programs (air quality, noise, marine water and sediment), interpret results against project and regulatory requirements, and prepare periodic environmental reports for clients and authorities. I also review contractors' environmental performance, support incident investigations and lessons-learned communication, contribute to JSA and risk assessments, and deliver training on environmental topics to project staff. I am aligning my work with ISO 14001 environmental management system practices

- Lead environmental modelling inputs to EIAs for coastal and offshore projects in the Arabian Gulf.
- Develop and implement project-specific Environmental and Waste Management Plans.
- Maintain environmental aspect/impact registers and link them to mitigation and monitoring.
- Evaluate contractor environmental performance and follow up on non-conformances.
- Support investigation of environmental incidents and communicate lessons learned.
- Provide environmental and climate-risk inputs to JSAs and deliver staff training.

**TECHNICAL SKILLS: Hydrodynamic and ocean circulation modeling using** MIKE, DELFT3D, TELEMAC, ADCIRC, and **Regional Climate Modeling** using WRF, Atmospheric modeling and air quality using WRF-Chem, CALPUFF, AERMOD and have strong programming skills in PYTHON, FORTRAN, and MATLAB.

## PROFESSIONAL EXPERIENCE

**King Fahd University of Petroleum & Minerals (KFUPM)**
***Applied Research Center for Environment and Marine Studies (ARCEMS)*** Dhahran, Saudi Arabia
*Research Engineer III* *April 2023 – Present*
*Role: Team Lead of Environmental Modeling Group*

- **Climate & Environmental Modeling Leadership:** Plan, design, and execute numerical modeling simulations for client-funded environmental impact assessment (EIA) and climate-related projects, including hydrodynamics, oil spill modeling, air dispersion, hydrology, noise pollution, water quality, and sediment transport modeling.
- **Atmospheric & Ocean Modeling Expertise:** Apply advanced modeling techniques using WRF (Weather Research and Forecasting), MIKE, ADCIRC, DELFT3D, SWAN, CALPUFF, AERMOD, and CadnaA for regional climate analysis, extreme weather events, and coastal dynamics.
- **Regional Climate Dynamics and Extremes in the Arabian Peninsula:** Conducted regional sea level rise projections, analyzed Rossby wave events and climate extremes in the Arabian Peninsula, and investigated teleconnections with ENSO and other climate indices to assess impacts on regional climate variability.

**National Centre for Sustainable Coastal Management** CHENNAI / INDIA
*Scientific Consultant* *January 2022 – March 2023*

**Project 1**-Assessment of Coastal Erosion and recommendation for suitable Interventions Green Coastal Infrastructure

- Objective to develop living shoreline infrastructure at two suitable locations of highly eroding coast along the Indian coast.
- The study involves identification of highly eroding sites and its modelling investigations (MIKE, GENCADE) involving design of green coastal protection for erosion control along with the cost benefit analysis.

**Project 2**- GCRF –South Asia Nitrogen Hub Project (SA-NH)

- To understand and quantify the complex system and impacts of nutrient sources and demands under various scenarios on lives and livelihoods in SA.
- Building on partner modelling capabilities (e.g. MIKE-SHE, WRF)

## LEADERSHIP EXPERIENCE

**Institute Women Council** Kharagpur, West Bengal / India
*Women Representative* *June 2018 – May 2019*

- Served as an elected member of student representative panel headed by director of institute.
- Suggestions and recommendations for the women's welfare and peaceful study and life inside campus
- Make sure each complaint receiving on the desk is attended and resolved in due time.

# JIYA ALBERT

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## PUBLICATION IN BOOK CHAPTERS

Climate Change and the Vulnerable Indian Coast. New Delhi, India: Ministry of Environment, Forestry and Climate Change: Book chapter #3 titled 'Tropical cyclogenesis for North Indian Ocean region'
*Prasad K. Bhaskaran, Jiya Albert, and Bishnupriya Sahoo*
*2018*

Climate Change Impacts on Water Resources: Hydraulics, Water Resources and Coastal Engineering: Book Chapter # 23 titled 'Seasonal and Inter-Annual Variability of Sea Surface Temperature and Its Correlation with Maximum Sustained Wind Speed in Bay of Bengal'
*Jiya Albert, Prasad K Bhaskaran*
*2021*
Extreme natural events: sustainable solutions for developing countries: Bool Chapter #5 titled "Extreme wind-Wave characteristics in the North Indian Ocean in a changing climate"
*Prasad K Bhaskaran, S Neelamani, Khaled Al-Salem, Athira Krishnan, Jiya Albert, S Sreelakshmi*
*2022*- Springer Nature Singapore

## PUBLICATION IN PEER-REVIEWED JOURNALS

1. **Albert, J**.; Gulakaram, V.S.; Vissa, N.K.; Bhaskaran, P.K.; Dash, M.K [2023]. Recent Warming Trends in the Arabian Sea: Causative Factors and Physical Mechanisms. *Climate*, *11*, 35. https://doi.org/10.3390/cli11020035
2. Shafeeque, M., Saleem, A. A., Chaitanya, A. V., Lateef, S., & **Albert, J*[Corresponding Author]**, [2025]. Chlorophyll-a Modulation in the Arabian Gulf Using Two-decades Merged Ocean-Colour Data, *Frontiers in Marine Science - Ocean Observation* ,12,1619821, https://doi.org/10.3389/fmars.2025.1619821
3. Neelavannan, K, Abdurahiman P, Jinoy, G., Rajeesh,M et al., **Albert, J** and Luai Alhems [2025], Nature Scientific Reports, 15, 15964 (2025). https://doi.org/10.1038/s41598-025-98136-3
4. Abdul Azeez S, A V. S. Chaitanya et al., **Albert, J*[Corresponding Author]** [2026], Sediment transport modelling of trenching activities: A case study for the Safaniya coastal waters in the Arabian Gulf, *Ocean Dynamics* 17 November 2025, https://doi.org/10.1007/s10236-025-01754-8
5. Saleem, A. V., Lateef A., Chaitanya, Shafeeque, **Albert, J*[Corresponding Author]** [2026], Advancing Marine Oil Spill detection in the Arabian Gulf: A Synergistic approach using SAR and Optical Satellite Imagery, Arabian Journal for Science and Engineering 17 December 2025, https://doi.org/10.1007/s13369-025-11036-6
6. **Albert, J**., Sahoo, B. and Bhaskaran, P.K. [2021]. Tropical cyclogenesis identification using eddy detection technique for the Bay of Bengal Basin. *Atmospheric Research*, 260,8095, https://doi.org/10.1016/j.atmosres.2021.105670.
7. **Albert, J**., Athira Krishnan, Bhaskaran, P.K. and Singh, K.S. [2022]. Role and influence of key atmospheric parameters in large-scale environmental flow associated with tropical cyclogenesis and ENSO in the North Indian Ocean basin. *Climate Dynamics*, https://doi.org/10.1007/s00382-021-05885-8
8. Kuvar SS, Ambily T, K Thatiparthi, Reshma,M, **Albert,J** Subbareddy B, Bhaskaran, P.K. [2022], Prediction of rapid intensification for land-falling extremely severe cyclonic storms in the Bay of Bengal, *Theoretical Applied Climatology* **147**, 1359–1377 (2022). https://doi.org/10.1007/s00704-022-03923-x
9. **Albert, J**., and Bhaskaran, P.K. [2020]. Ocean heat content and its role in tropical cyclogenesis for the Bay of Bengal basin. *Climate Dynamics*, 55(11), 3343-3362. https://doi.org/10.1007/s00382-020-05450-9
10. **Albert, J**., and Bhaskaran, P.K. [2020]. Evaluation of multi-level upper ocean heat content and its relationship with intensity and translation speed of tropical cyclones in the North Indian Ocean. *Journal of Theoretical and Applied Climatology*, 154,3, 1333–1342 (2023). https://doi.org/10.1007/s00704-023-04641-8
11. **Albert, J**., and Bhaskaran, P.K. [2020]. Optimal grid resolution for the detection lead time of cyclogenesis in the North Indian Ocean. *Journal of Atmospheric and Sola and Terrestrial Physics*, 204, https://doi.org/10.1016/j.jastp.2020.105289
12. **Albert, J**., and Bhaskaran, P.K. [2020]. Evaluation of track length, residence time and translational speed for tropical cyclones in the NIO. *ISH Journal of Hydraulic Engineering*, https://doi.org/10.1080/09715010.2020.1825124
13. Singh, K.S., **Albert, J**., Bhaskaran, P.K. and Alam, P. [2021]. Assessment of extremely severe cyclonic storms over Bay of Bengal and performance evaluation of ARW model in the prediction of track and intensity. *Theoretical and Applied Climatology*, 143(3),1181- 1194. https://doi.org/10.1007/s00704-020-03510-y

14. Singh, K.S., **Albert, J**., Bhaskaran, P.K. and Alam, P. [2021]. Numerical simulation of an extremely severe cyclonic storm over the Bay of Bengal using WRF modelling system: influence of model initial condition. *Modeling Earth Systems and Environment*, https://doi.org/10.1007/s40808-020-01069-1
15. Singh, K.S., Alam, P., **Albert, J**. and Bhaskaran, P.K. [2020]. Numerical simulation of an extremely severe cyclonic storm Hudhud over the North Indian Ocean in a medium range scale: influence of cloud microphysical schemes. *Pure and Applied Geophysics*, 177(12), 5895-5910. https://doi.org/10.1007/s00024-020-02596-9

## PUBLICATION IN PEER-REVIEWED CONFERENCES

1. Muhammad Shafeeque, Abdul Azeez Saleem, Akurathi VS Chaitanya, Salim Lateef, Jiya Albert, Luai Muhammad Alhems[2025]. Decreasing Chlorophyll-a Trend in the Arabian Gulf and Associated Climate Variables, Fall Meeting of the Americal Geophysical Union(AGU),17 December 2025,
2. J Albert Muhammad Shafeeque, Vadakke-Chanat Sayoob, S Abdul Azeez, Sai Chaitanya Akurathi Venkata, Lateef Salim[2024]. Mapping bathymetry and shallow water benthic habitats in the Arabian Gulf using multispectral Sentinel-2 Satellite imagery, Proceedings of MedGU-24, 11 November 2024, Springer IEREK ASTI Book Series

## EDUCATION & OTHER

**INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR** *PHD-GRADUATION YEAR-2021*

**COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY** *MTECH-GRADUATION YEAR-2017*

**MAHATMA GANDHI UNIVERSITY KOTTAYAM** *BE Civil Eng- GRADUATION YEAR-2014*

**LANGUAGES:** English (fluent, CEFR-C2), Malayalam (native), Hindi (basic)

**OTHER EDUCATION:** Gold Medalist from Cochin University (CUSAT) in Master of Technology - Ocean Technology

**INTERESTS:** Indian Classical Dance, Music, Cooking